CS190C Assignment 2 Report

1. Experimental Setup

This assignment studies how small single-GPU pilot runs can guide a larger multi-GPU training decision. I trained LLaMA-style decoder-only language models from scratch on the official `roneneldan/TinyStories` training split and used the official validation split for model selection and final reporting. The tokenizer was `roneneldan/TinyStories-33M`, with vocabulary size 50,257 and EOS reused as the padding token.

All required runs used sequence length 512, AdamW, cosine learning-rate decay with warmup, weight decay 0.1, and seed 42. I used Hugging Face `transformers` for `LlamaForCausalLM` and used `accelerate` directly for single-GPU and two-GPU training. TensorBoard logging and checkpointing were enabled in the training loop.

1.1 System and Software Setup

The experiments were run on a `H20` cluster node through Hugging Face `accelerate`. Single-GPU pilot experiments used `accelerate_configs/single_gpu.yaml`, which sets `distributed_type: "NO"`, `num_processes: 1`, and `mixed_precision: bf16`. The final scale-up experiment used `accelerate_configs/two_gpu_ddp.yaml`, which sets `distributed_type: MULTI_GPU`, `num_processes: 2`, and `mixed_precision: bf16`.

The software stack included PyTorch, Hugging Face `transformers`, `accelerate`, `datasets`, TensorBoard, PyYAML, tqdm, safetensors, pandas, and matplotlib. The dataset was loaded through the Hugging Face `datasets` cache, and checkpoints were saved with `accelerator.save_state(...)`, including model weights, optimizer state, scheduler state, and random states.

The pilot effective batch size was:

Stage	GPUs	Per-device batch	Gradient accumulation	Effective batch	Tokens / step
Pilot	1	8	4	32 sequences	16,384
Scale-up	2	8	8	128 sequences	65,536

2. Pilot Experiments

I ran six single-GPU pilot experiments. The first three form a learning-rate sweep on model S. After selecting the best learning rate from that sweep, I ran a model-size sweep over XS, S, and M.

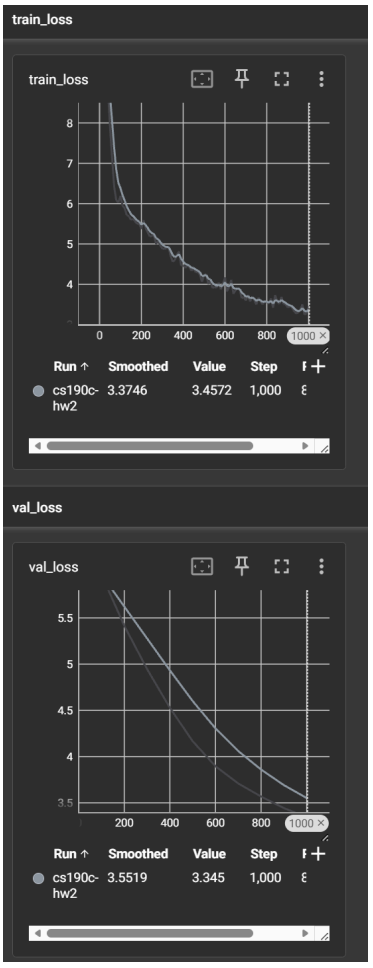
2.1 Learning-Rate Sweep on Model S

Run	Model	Learning rate	Steps	Tokens seen	Final validation loss	Final perplexity
`pilot_s_lr2e4`	S	0.0002	1000	16,384,000	3.5246	33.9412

Run	Model	Learning rate	Steps	Tokens seen	Final validation loss	Final perplexity
`pilot_s_lr5e4`	S	0.0005	1000	16,384,000	3.3941	29.7870
`pilot_s_lr1e3`	S	0.0010	1000	16,384,000	3.3450	28.3618

The learning-rate sweep showed that 0.001 gave the best validation loss among the three tested values. The lower learning rates were stable but learned more slowly within the fixed 1000-step budget. Based on this result, I used 0.001 for the size sweep.

- **TensorBoard curve for the best S learning-rate pilot run:**



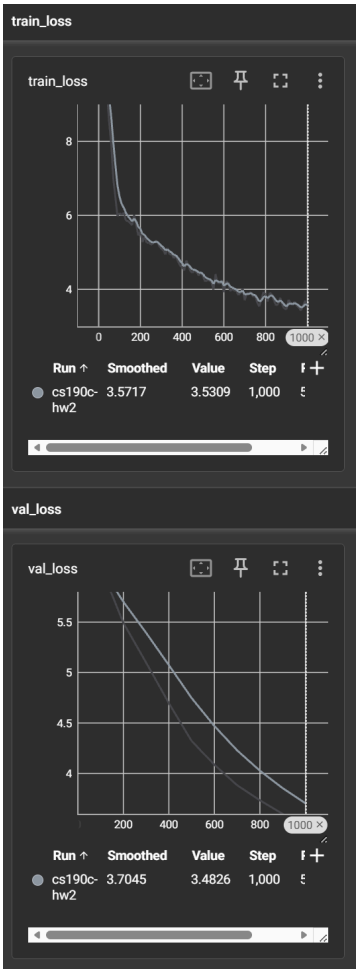
2.2 Model-Size Sweep at the Best Learning Rate (lr = 1e-3)

Run	Model config	Learning rate	Steps	Tokens seen	Final validation loss	Final perplexity
`pilot_xs_bestlr`	XS	0.0010	1000	16,384,000	3.4826	32.5450
`pilot_s_bestlr`	S	0.0010	1000	16,384,000	3.3450	28.3618
`pilot_m_bestlr`	M	0.0010	1000	16,384,000	3.2912	26.8747

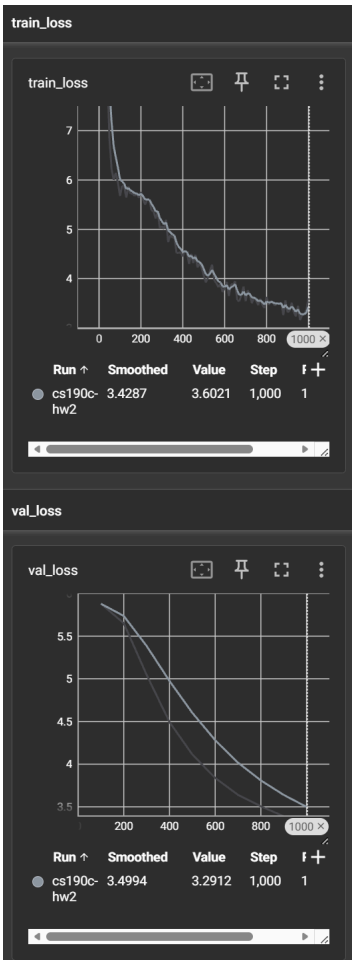
The size sweep showed a clear trend: larger models achieved lower validation loss under the same token budget and optimizer settings. XS was the weakest, S improved over XS, and M achieved the best validation loss. This suggested that the training setup was not yet saturated by model capacity and that scaling to M was a reasonable final choice.

The extra `pilot_s_bestlr` run reached a similar validation-loss range as the earlier `pilot_s_lr1e3` run, though it did not exactly reproduce the earlier best value. This is reasonable for short 1000-step pilot runs, where data order, initialization, and GPU nondeterminism can create visible run-to-run variation. The main conclusion remains unchanged because model M achieved the best validation loss in the size sweep.

- **TensorBoard curve for the XS model-size pilot run:**



- **TensorBoard curve for the M model-size pilot run:**



3. Scale-Up Decision

For the final two-GPU run, I selected model M with learning rate 0.001. This choice was based on two observations from the pilot study:

1. The learning-rate sweep showed that 0.001 was the best tested learning rate for model S.
2. The model-size sweep showed that M outperformed XS and S under the same pilot token budget.

Compared with the pilot M run, the final run increased three required dimensions:

Quantity	Pilot M	Final M scale-up
GPUs	1	2
Effective batch size	32 sequences	128 sequences
Training tokens	16,384,000	196,608,000
Training steps	1000	3000

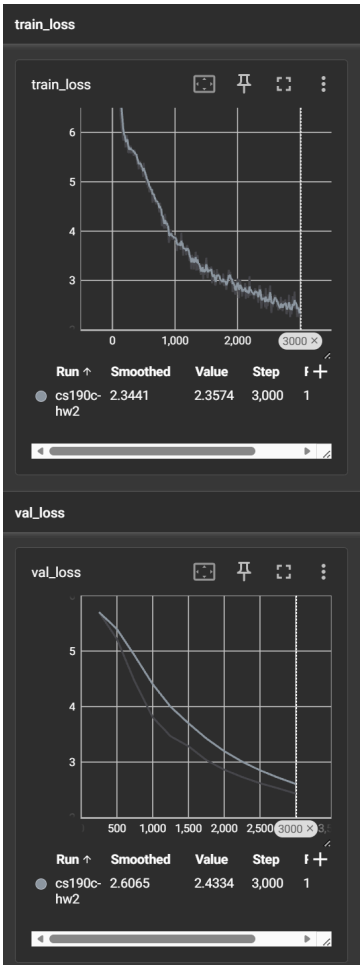
This satisfies the assignment requirement to increase at least two of GPU count, model size, total tokens, and effective batch size. I kept the model family, tokenizer, sequence length, optimizer, and learning-rate schedule fixed so that the scale-up comparison remained controlled.

4. Final Validation Results

Run	Model	Accelerate config	GPUs	Effective batch	Tokens seen	Final validation loss	Final perplexity
`scaleup_m`	M	`two_gpu_ddp.yaml`	2	128 sequences	196,608,000	2.4334	11.3970

The final run substantially improved over the pilot M run, reducing validation loss from 3.2912 to 2.4334 and perplexity from 26.8747 to 11.3970. This behavior matched the expectation from the pilot results: model M was the strongest pilot model, and giving it more training tokens and a larger effective batch on two GPUs improved validation performance.

- **TensorBoard curve for the final two-GPU scale-up run:**



The TensorBoard logs for Part C are saved under:

```
outputs/scaleup_m/cs190c-hw2/
```

The exported TensorBoard figures are saved in:

```
plots/
```

Relevant figures include `plots/scaleup_m.png` for the final run and `plots/pilot_xs_bestlr.png` and `plots/pilot_m_bestlr.png` for the size sweep.

5. Discussion

The pilot experiments were useful for making the final training decision. The learning-rate sweep reduced uncertainty about the optimizer setting, and the size sweep showed that larger models still benefited from the fixed pilot token budget. Because M was the best pilot model, it was the natural candidate for the two-GPU run.

The scale-up behaved as expected. Increasing total tokens from 16.4M to 196.6M produced a much lower validation loss, and using two GPUs allowed a larger effective batch size while keeping the run practical. The final result is not just a lower loss; it is also supported by a controlled sequence of smaller experiments.

I did not attempt the optional FSDP or DeepSpeed bonus experiments. The final distributed run used standard two-GPU DDP through `accelerate`.

6. Submitted Artifacts

The submission includes:

- Completed training and evaluation code in `scripts/train.py` and `scripts/evaluate.py`.
- Pilot results table in `results/pilot_results.csv`.
- Final scale-up results in `results/scaleup_results.csv`.
- Part C TensorBoard logs in `outputs/scaleup_m/cs190c-hw2/`.
- Exported TensorBoard figures in `plots/`.